

Tide retrospective: the multivariate opener (H-recovery programme, stages H0–H1)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

With the one-dimensional Theorem 3.1 complete at every order, the note’s remaining commission is the multivariate statement — also the Susceptibility Primer’s missing Stage 3. The scoping consult (archived in the tide log, with the note attached) made four rulings that shape the whole programme: recover the Hessian first, before any jet package, since H -recovery alone forces every new foundation; represent the ambient space as EuclideanSpace $\mathbb{R}$ (Fin d); build *one* project-local dilation wrapper so downstream proofs never touch Mathlib’s change-of-variables surface; and — the structural prize — handle the eventual degree- k tensor injectivity by covariance-Gram positive-definiteness from the Gaussian’s full support, the same open-support argument as one dimension, avoiding Wick/Isserlis entirely.

2 The thing formalised

Three declarations in `Laplace/Multi/Dilation.lean`¹: the ambient abbreviation, the dilation wrapper $\int f = q^d \int f(q \cdot x)$ for $q > 0$, and its indicator form for the expanding domains of the later dominated-convergence stage. The consult’s named trap — the pushforward carries q^{-d} while the substitution carries q^d — is fixed here once, and the orientation was checked numerically first (the $d = 2$ Gaussian-with-cubic integral and a box indicator, both matching to the printed precision).

3 Notes

The only friction was the catalogued `Measure.`-namespace gotcha, striking for `integral_comp_smul` exactly as the catalogue predicts for its 1D sibling. Stages H2–H6 follow: the Gaussian package via whitening (the flagged API spike — Mathlib’s multivariate Gaussian infrastructure should be surveyed first), the local quadratic rescaling, the expanding-domain dominated convergence, normalization, and the pairwise Hessian recovery.

¹`EuclidD, integral_dilation, setIntegral.dilation.`